

Core Notes on Ordinary Least Squares

Estimation, Uncertainty, Statistical Inference, Causality, and Gauss–Markov

Introductory Econometrics

Road map

The central question is: given the linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$, what does OLS estimate, how uncertain is it, and when can we interpret it causally? We proceed from the estimator to its variance, estimate the unknown error variance, conduct inference, and finally establish the Gauss–Markov result.

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1 The linear regression model

For observations $i = 1, \dots, n$, suppose

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + u_i.$$

In matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where $\mathbf{y}$ is $n \times 1$, $\mathbf{X}$ is $n \times p$, $\boldsymbol{\beta}$ is $p \times 1$, and $\mathbf{u}$ is $n \times 1$. Here $p = k + 1$ if the model contains an intercept.

The parameter vector $\boldsymbol{\beta}$ describes the population linear relationship. The error u_i contains all determinants of y_i not explicitly included in $\mathbf{X}$.

Key assumptions used in these notes

1. **Linear model:** $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$.
2. **Full column rank:** $\text{rank}(\mathbf{X}) = p$, so $(\mathbf{X}'\mathbf{X})^{-1}$ exists.
3. **Zero conditional mean:** $\mathbb{E}(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}$.
4. **Homoskedastic, uncorrelated errors:** $\text{Var}(\mathbf{u} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$.
5. **Normality** (only for exact small-sample inference): $\mathbf{u} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

Assumptions 1–3 deliver conditional unbiasedness. Assumption 4 gives the familiar OLS variance formula and Gauss–Markov efficiency. Assumption 5 is stronger and is not required for OLS to be unbiased.

2 Deriving the OLS estimator

Here $\boldsymbol{\beta}$ is the true population parameter, $\mathbf{b}$ is an arbitrary candidate coefficient vector, and $\hat{\boldsymbol{\beta}}$ is the particular value of $\mathbf{b}$ that minimizes the sum of squared residuals. Thus,

$$\hat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b}} (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}).$$

To solve this problem, write the objective function for a generic candidate $\mathbf{b}$:

$$SSR(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}).$$

Expanding and differentiating,

$$\begin{aligned} SSR(\mathbf{b}) &= \mathbf{y}'\mathbf{y} - 2\mathbf{b}'\mathbf{X}'\mathbf{y} + \mathbf{b}'\mathbf{X}'\mathbf{X}\mathbf{b}, \\ \frac{\partial SSR(\mathbf{b})}{\partial \mathbf{b}} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\mathbf{b}. \end{aligned}$$

Setting the derivative equal to zero gives the *normal equations*,

$$\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0},$$

and therefore

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Substituting the population model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$ gives the most useful identity in OLS:

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \\ &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}. \end{aligned}$$

Hence

$$\widehat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}.$$

What creates estimation error?

The gap between $\widehat{\beta}$ and β is the part of the unobserved error $\mathbf{u}$ that happens to align with the regressors in this sample. More variation in $\mathbf{X}$ makes $(\mathbf{X}'\mathbf{X})^{-1}$ smaller and generally improves precision.

3 Exogeneity

To study whether OLS is centered on the true parameter, we need an assumption about the relationship between the regressors and the unobserved error. The *zero conditional mean* (or exogeneity) assumption is

$$\mathbb{E}(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}.$$

It says that, once we condition on the observed regressors, the remaining unobserved determinants of $\mathbf{y}$ have mean zero. Equivalently,

$$\mathbb{E}(\mathbf{y} \mid \mathbf{X}) = \mathbf{X}\beta.$$

Exogeneity is not a mechanical property of OLS

The sample normal equations imply $\mathbf{X}'\widehat{\mathbf{u}} = \mathbf{0}$ by construction; this is always true for fitted OLS residuals. In contrast, $\mathbb{E}(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}$ concerns the unobserved population error and is an identifying assumption that must be justified from the research design or economic setting.

4 Unbiasedness and the variance of OLS

Conditioning on the observed regressor matrix $\mathbf{X}$,

$$\begin{aligned}\mathbb{E}(\widehat{\beta} \mid \mathbf{X}) &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}(\mathbf{u} \mid \mathbf{X}) \\ &= \beta.\end{aligned}$$

Thus, under zero conditional mean, OLS is conditionally unbiased.

For the conditional variance, use the identity $\text{Var}(\mathbf{A}\mathbf{z} \mid \mathbf{X}) = \mathbf{A} \text{Var}(\mathbf{z} \mid \mathbf{X}) \mathbf{A}'$:

$$\begin{aligned}\text{Var}(\widehat{\beta} \mid \mathbf{X}) &= \text{Var}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \mid \mathbf{X}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{Var}(\mathbf{u} \mid \mathbf{X}) \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.\end{aligned}$$

Under $\text{Var}(\mathbf{u} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$,

$$\text{Var}(\widehat{\beta} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}.$$

The variance of coefficient $\widehat{\beta}_j$ is the j th diagonal element:

$$\text{Var}(\widehat{\beta}_j \mid \mathbf{X}) = \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}.$$

If errors are heteroskedastic

When $\text{Var}(\mathbf{u} \mid \mathbf{X}) = \mathbf{\Omega} \neq \sigma^2 \mathbf{I}_n$, the correct conditional variance is

$$(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.$$

The OLS coefficients can remain unbiased, but the homoskedastic standard errors are generally wrong. This motivates heteroskedasticity-robust standard errors.

5 Residuals and estimating σ^2

Define the projection and residual-maker matrices

$$\mathbf{P}_X = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad \mathbf{M}_X = \mathbf{I}_n - \mathbf{P}_X.$$

The fitted values and residuals are

$$\hat{\mathbf{y}} = \mathbf{P}_X \mathbf{y}, \quad \hat{\mathbf{u}} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{M}_X \mathbf{y}.$$

Since $\mathbf{M}_X \mathbf{X} = \mathbf{0}$,

$$\hat{\mathbf{u}} = \mathbf{M}_X(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) = \mathbf{M}_X \mathbf{u}.$$

The residual sum of squares is therefore

$$SSR = \hat{\mathbf{u}}'\hat{\mathbf{u}} = \mathbf{u}'\mathbf{M}_X \mathbf{u},$$

because $\mathbf{M}_X$ is symmetric and idempotent: $\mathbf{M}_X' = \mathbf{M}_X$ and $\mathbf{M}_X^2 = \mathbf{M}_X$.

5.1 Why divide by $n - p$ rather than n ?

Under homoskedasticity, the expected residual sum of squares is

$$\mathbb{E}(SSR \mid \mathbf{X}) = (n - p)\sigma^2.^1$$

Therefore, an unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{SSR}{n - p} = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n - p}.$$

Degrees of freedom

The n raw errors contain n directions of variation. Fitting p coefficients uses up p directions, leaving $n - p$ independent residual directions. With an intercept and k slopes, $p = k + 1$ and the denominator is $n - k - 1$.

The estimated covariance matrix of OLS is

$$\widehat{\text{Var}}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1},$$

so

$$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}.$$

¹Since $SSR = \mathbf{u}'\mathbf{M}_X \mathbf{u}$ and, for a mean-zero vector $\mathbf{z}$, $\mathbb{E}(\mathbf{z}'\mathbf{A}\mathbf{z}) = \text{tr}[\mathbf{A} \text{Var}(\mathbf{z})]$, we have $\mathbb{E}(SSR \mid \mathbf{X}) = \text{tr}(\mathbf{M}_X \sigma^2 \mathbf{I}_n) = \sigma^2 \text{tr}(\mathbf{M}_X)$. Moreover, using the cyclic trace identity $\text{tr}(\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{A})$, $\text{tr}(\mathbf{M}_X) = \text{tr}(\mathbf{I}_n) - \text{tr}[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] = n - \text{tr}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}] = n - \text{tr}(\mathbf{I}_p) = n - p$.

6 Statistical inference

If $\mathbf{u} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, then a linear transformation of $\mathbf{u}$ is also normal:

$$\hat{\boldsymbol{\beta}} \mid \mathbf{X} \sim N(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}).$$

For coefficient j ,

$$\hat{\beta}_j \mid \mathbf{X} \sim N(\beta_j, \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}).$$

Normality of the errors gives this result exactly in finite samples. More generally, under standard regularity conditions, the central limit theorem implies that as $n \rightarrow \infty$, the standardized OLS estimator converges in distribution to a standard normal random variable:

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \xrightarrow{d} N(0, 1).$$

Thus, under the null hypothesis $H_0 : \beta_j = \beta_{j,0}$, we use

$$t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)}.$$

For large n , t_j is approximately $N(0, 1)$. Under normally distributed errors, it has the exact finite-sample t_{n-p} distribution.

For a two-sided test at significance level α , reject H_0 when

$$|t_j| > t_{1-\alpha/2, n-p}.$$

A $(1 - \alpha) \times 100\%$ confidence interval is

$$\hat{\beta}_j \pm t_{1-\alpha/2, n-p} SE(\hat{\beta}_j).$$

Three objects that students often confuse

Object	Meaning
β_j	Fixed but unknown population parameter
$\hat{\beta}_j$	Random estimator before sampling; a realized number after observing the sample
$SE(\hat{\beta}_j)$	Estimated sampling uncertainty of the estimator, not the standard deviation of y

7 Statistical inference versus causal inference

These are different questions.

Statistical inference	Causal inference
How precisely have we estimated the population regression coefficient?	What would happen to Y if we intervened and changed X ?
Uses standard errors, confidence intervals, and hypothesis tests.	Requires an identification argument linking variation in X to a counterfactual comparison.
Answers whether sampling uncertainty is small.	Answers whether the coefficient has a causal interpretation.

Consider the population equation

$$Y = \beta_0 + \beta_1 X + U.$$

If

$$\mathbb{E}(U \mid X) = 0,$$

then units with different values of X do not differ systematically in unobserved determinants of Y . Therefore,

$$\mathbb{E}(Y \mid X = x) = \beta_0 + \beta_1 x,$$

and, under a valid structural interpretation of the equation, β_1 can represent the causal effect of changing X by one unit.

A small standard error does not establish causality

One can estimate a confounded association extremely precisely. Statistical significance concerns uncertainty around the estimated association; causal identification concerns whether that association equals the effect of an intervention.

7.1 Example: education and wages

Suppose

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \text{education}_i + u_i,$$

where u_i includes motivation, family background, and ability. If ability raises both schooling and wages, then

$$\mathbb{E}(u_i \mid \text{education}_i = 1) \neq \mathbb{E}(u_i \mid \text{education}_i = 0).$$

The OLS coefficient combines the return to education with selection on unobserved ability. More observations may reduce the standard error, but they do not remove this omitted-variable bias.

In the simple regression with an intercept, the weaker condition $\text{Cov}(X, U) = 0$ is enough for unbiasedness of the OLS slope. The zero conditional mean assumption $\mathbb{E}(U \mid X) = 0$ implies $\text{Cov}(X, U) = 0$, but the reverse need not hold: a nonlinear relationship between X and U can have zero covariance.

8 Gauss–Markov: why OLS is BLUE

Under linearity, full rank, zero conditional mean, and spherical errors, OLS is the **Best Linear Unbiased Estimator** (BLUE). “Best” means smallest conditional variance among all estimators that are linear in $\mathbf{y}$ and unbiased.

Consider any linear estimator

$$\tilde{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y},$$

where $\mathbf{A}$ is a $p \times n$ matrix that may depend on $\mathbf{X}$ but not on $\mathbf{y}$. Its conditional expectation is

$$\begin{aligned}\mathbb{E}(\tilde{\boldsymbol{\beta}} \mid \mathbf{X}) &= \mathbf{A}\mathbb{E}(\mathbf{y} \mid \mathbf{X}) \\ &= \mathbf{A}\mathbf{X}\boldsymbol{\beta}.\end{aligned}$$

For this to equal $\boldsymbol{\beta}$ for every possible $\boldsymbol{\beta}$, we must have

$$\boxed{\mathbf{A}\mathbf{X} = \mathbf{I}_p}.$$

The OLS matrix is

$$\mathbf{A}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}',$$

and indeed $\mathbf{A}_{OLS}\mathbf{X} = \mathbf{I}_p$. Any other linear unbiased estimator can be written

$$\mathbf{A} = \mathbf{A}_{OLS} + \mathbf{D},$$

where unbiasedness implies

$$(\mathbf{A} - \mathbf{A}_{OLS})\mathbf{X} = \mathbf{D}\mathbf{X} = \mathbf{0}.$$

Thus

$$\tilde{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y} = (\mathbf{A}_{OLS} + \mathbf{D})\mathbf{y} = \hat{\boldsymbol{\beta}} + \mathbf{D}\mathbf{y}.$$

Now compare conditional variances. Under $\text{Var}(\mathbf{y} \mid \mathbf{X}) = \sigma^2\mathbf{I}_n$,

$$\begin{aligned}\text{Var}(\tilde{\boldsymbol{\beta}} \mid \mathbf{X}) &= \sigma^2(\mathbf{A}_{OLS} + \mathbf{D})(\mathbf{A}_{OLS} + \mathbf{D})' \\ &= \sigma^2\mathbf{A}_{OLS}\mathbf{A}_{OLS}' + \sigma^2\mathbf{D}\mathbf{D}' + \sigma^2\mathbf{A}_{OLS}\mathbf{D}' + \sigma^2\mathbf{D}\mathbf{A}_{OLS}'.\end{aligned}$$

The cross terms are zero because

$$\mathbf{A}_{OLS}\mathbf{D}' = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{D}\mathbf{X})' = \mathbf{0}.$$

Therefore,

$$\boxed{\text{Var}(\tilde{\boldsymbol{\beta}} \mid \mathbf{X}) - \text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2\mathbf{D}\mathbf{D}' \succeq \mathbf{0}}.$$

The notation $\succeq \mathbf{0}$ means positive semidefinite: for any vector $\mathbf{c}$,

$$\mathbf{c}'\mathbf{D}\mathbf{D}'\mathbf{c} = \|\mathbf{D}'\mathbf{c}\|^2 \geq 0.$$

Hence no other linear unbiased estimator has a smaller variance for any linear combination of the coefficients.

The idea behind BLUE

Every linear unbiased alternative equals OLS plus an extra term $D\mathbf{y}$. The restriction $D\mathbf{X} = \mathbf{0}$ prevents this term from changing the estimator's mean, but its variance $\sigma^2 D D'$ is nonnegative. It can add noise; it cannot improve precision.

What Gauss–Markov does not say

It does not say that OLS is best among *all* possible estimators, that OLS is normally distributed, or that OLS coefficients are causal. It compares only linear unbiased estimators under the stated variance assumptions.

9 A compact formula sheet

$$\text{Model: } \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}.$$

$$\text{OLS: } \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

$$\text{Estimation error: } \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}.$$

$$\text{Unbiasedness: } \mathbb{E}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \boldsymbol{\beta} \quad \text{if } \mathbb{E}(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}.$$

$$\text{Variance: } \text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

$$\text{Residuals: } \hat{\mathbf{u}} = \mathbf{M}_X \mathbf{y}, \quad \mathbf{M}_X = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'.$$

$$\text{Error variance: } \hat{\sigma}^2 = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n-p}.$$

$$\text{Estimated variance: } \widehat{\text{Var}}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}.$$

$$t \text{ statistic: } t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)}.$$

$$\text{Confidence interval: } \hat{\beta}_j \pm t_{1-\alpha/2, n-p} SE(\hat{\beta}_j).$$

$$\text{Gauss–Markov: } \text{Var}(\tilde{\boldsymbol{\beta}} \mid \mathbf{X}) - \text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2 D D' \succeq \mathbf{0}.$$

10 Quick self-check

1. Why does $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ depend on $\mathbf{X}'\mathbf{u}$?
2. Which assumption shows unbiasedness? Which additional assumption gives the standard OLS variance formula?
3. Why is the residual-variance estimator divided by $n - p$?
4. Can an estimate be statistically significant but not causal? Explain.
5. In the Gauss–Markov proof, why does unbiasedness require $\mathbf{A}\mathbf{X} = \mathbf{I}_p$?
6. Why is $D D'$ always positive semidefinite?